

Question	Answer	Marks
1(a)	base units: $\text{kg m s}^{-2} \times \text{m}$ $= \text{kg m}^2 \text{s}^{-2}$	A1
1(b)(i)	distance of COG from P (= GP) $= 17 \cos 45^\circ - 4.0$ or $(144.5)^{1/2} - 4.0$ (= 8.0 cm)	C1
	moment = $0.15 \times 8.0 \times 10^{-2}$ $= 1.2 \times 10^{-2} \text{ N m}$	A1
1(b)(ii)	(line of action of) weight acts through pivot/P or distance between (line of action of) weight and pivot/P is zero	B1
	(so) weight does not have a moment about pivot/P	B1